Fingerprinting Language-Model Weights via the Distributional Divergence of Transferable Low-Rank Probes

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Abstract

Determining whether two language models have (nearly) identical weights is central to model-copy detection, provenance attribution, and adversarial weight-tampering auditing, but is difficult because hidden states are inaccessible in production and because two models can produce identical outputs while differing in their parameters. We introduce a two-stage approach built on an observable, distribution-level signal. First, we generate *representation-aligned probes*: sentences the model's own intermediate representation places near a common reference, so the probe set stays low-rank and concentrated across models, though the sentences differ on the surface. They are produced by a HotFlip-style discrete optimization over a shared subword vocabulary, a one-time *white-box* step on a local open-weight model. Second, we measure their mean per-probe *distributional divergence* (KL) of the induced next-token distributions as a fingerprint. Detection is *black-box*, requiring only the top-k log-probabilities of the served model.

The fingerprint is a monotone, sensitive detector of weight perturbation. In a ground-truth-controlled experiment on a local open-weight model, the detector, reading only the top-20 log-probabilities of the served checkpoint, fires on a 250-step LoRA fine-tune ($D_{\mathrm{KL}} = 0.17$) and on a wholesale model swap ($D_{\mathrm{KL}} = 4.11$), an order of magnitude apart. Calibration under injected Gaussian noise of scale $r\sigma$ on three open-weight models (Qwen3.5-0.8B, Llama-3.2-1B, Gemma-3-1B) yields a divergence that grows quadratically in r, consistent with the Fisher-information prediction and stable across model families and the two Qwen layers ($L8/L16$). A zero-noise determinism check over the DeepSeek API confirms that the black-box readout is deterministic end to end against a live model. Across families, tokenizer mismatch drives the divergence to a saturated maximum, so a model swap is caught with a large margin ($D_{\mathrm{KL}} = 4.1$); within a family, a black-box statistic preserves the calibration at roughly half to three-quarters of the white-box magnitude. An ablation of the probe-construction role (§5.3) shows that representation alignment contributes a transferable, low-rank probe set, the property on which black-box cross-model comparison rests; it does not maximize the raw per-probe divergence.

1 Introduction

Large language models are expensive to train, and their weights are the primary carrier of that investment. This raises a natural and increasingly practical question: given two models, can we determine whether their parameters are (nearly) identical? The question matters for at least three reasons. First, *copy detection*: verifying whether a released checkpoint has been surreptitiously derived from a proprietary model. Second, *provenance*: attributing a model to its training lineage. Third, *tamper auditing*: detecting whether a served model has been silently modified, for instance, by a malicious actor injecting fine-tuned backdoors or by an operator swapping weights at a load balancer.

Answering this question is hard for two coupled reasons. On one hand, the weights themselves are typically not exposed: a deployment only offers token distributions (or, at most, top-k log-probabilities) through an API. On the other, *output equivalence does not imply weight equivalence*: two very different parameterizations can realize the same input–output mapping, so agreement on a held-out set of prompts is a necessary but weak signal.

We propose to sidestep both difficulties by exploiting the *second-order* response of the model's observable distribution: the distributional divergence of the observable next-token distribution, whose quadratic response is controlled by the Fisher information. A model's weights induce a specific local geometry in representation space. If we can produce a set of sentences that map to a tight cluster under the model's own intermediate representation, then the way in which these representation-aligned sentences spread in next-token probability space is a signature of the weights themselves. Perturbing the weights perturbs this distributional geometry in a way that a single-output comparison cannot reveal. Throughout, "alignment" is defined relative to the model's own representation: the probes must land on (nearly) the same latent representation, and are *not* required to be synonyms in any linguistic sense. The role of this alignment is structural: the probe set stays low-rank and concentrated when read on a different model, which is what makes the fingerprint transferable across models.

Concretely, we make three contributions:

1. **Constrained representation-aligned generation** (§4.1): a discrete optimization procedure that, given a reference sentence, emits multiple sentences the model's representation places near it, so the probe set is low-rank and transferable across models, while differing on the surface, using only tokens from a restricted (shared) subword vocabulary. The output is always a discrete token sequence, never a vector. Synthesis is a one-time white-box step performed on a local open-weight model; detection reads only top-k log-probabilities.

2. **Distributional-divergence fingerprinting** (§4.2): a fingerprint defined as the mean per-probe next-token KL divergence between two models, together with a two-sample change-detection test (mean per-probe D_{KL} as the primary verdict, with a per-token χ^2 aggregate plus mean-vector and covariance-spectrum diagnostics).

3. **Empirical calibration and limits** (§5): a ground-truth-controlled black-box readout that, reading only the top-k log-probabilities, detects fine-tuning, weight quantization, and wholesale model swapping by a wide margin (§5.4); a calibration curve showing that the fingerprint diverges monotonically with added Gaussian weight noise, consistent across model families and the two Qwen layers ($L8/L16$); a zero-noise determinism check over the DeepSeek API; and a characterization of cross-family comparison in which tokenizer mismatch drives the fingerprint to its maximal (saturated) signal, so a change of model family is detected decisively, while fine-grained attribution remains bounded.

2 Related Work

Discrete adversarial attacks. Our representation-aligned generator descends from the family of gradient-based discrete text optimizers initiated by Ebrahimi et al. [2018], which approximates the effect of a token substitution by the inner product of the input-embedding gradient and the candidate embedding. Subsequent work refined this idea with beam search and constrained candidate sets [Jin et al., 2020]. We differ in purpose and in our candidate constraint. We optimize for *preservation of the model's representation*, so the output must remain near the reference in the model's own representation, whereas HotFlip-style attacks seek a label flip. Our candidates are also drawn from a shared subword vocabulary, which is essential for cross-model comparison.

Representation similarity. Measuring the similarity of neural representations has a long history, with centered kernel alignment (CKA) [Kornblith et al., 2019] and representational similarity analysis [Kriegeskorte et al., 2008] as the canonical tools. These methods compare the *internal activations* of two networks and thus require white-box access. Our fingerprint is instead computed on the observable log-probability surface, making it compatible with API-only deployments, while still being grounded in the intermediate representation through the probe-construction step.

Model fingerprinting and extraction. Model extraction attacks [Tramèr et al., 2016] and watermarking [Kirchenbauer et al., 2023] both fingerprint a model's *behavior*; weight-recovery attacks [Rolnick and Kording, 2020] recover the *parameters* directly. Our work occupies the middle ground: we estimate a second-order statistic that is sensitive to the weights but does not require recovering them, and is observable without hidden states.

Query-based model fingerprinting. The closest prior art infers a model's origin by actively probing it with optimized queries [Chen et al., 2022]. We extend this line of work in two directions. First, our probes are *actively synthesized* to be representation-aligned, placed together in the model's own intermediate representation; the signal is therefore a within-representation-cluster distributional divergence. Second, our statistic is the *KL divergence* between the per-probe next-token distributions, which is far more sensitive to weight perturbation than their mean.

3 Problem Formulation

Let a language model M_θ be parameterized by weights $\theta \in \mathbb{R}^P$, with vocabulary $\mathcal{V}$ and input-embedding matrix $\boldsymbol{E} \in \mathbb{R}^{|\mathcal{V}| \times D}$. For an input token sequence $\boldsymbol{x} = (x_1, \ldots, x_T) \in \mathcal{V}^T$, denote by $\boldsymbol{h}^{(\ell)}(\boldsymbol{x}) \in \mathbb{R}^{T \times D}$ the hidden states at intermediate layer ℓ, and by $\bar{\boldsymbol{h}}^{(\ell)}(\boldsymbol{x}) = \frac{1}{T} \sum_t \boldsymbol{h}_t^{(\ell)}(\boldsymbol{x}) \in \mathbb{R}^D$ its mean-pooled sentence vector. The model maps a prefix to a next-token distribution $p_\theta(\cdot \mid \boldsymbol{x}) = \mathrm{softmax}(\boldsymbol{W}\boldsymbol{h}^{(L)}(\boldsymbol{x}))$ over the vocabulary.

Goal. Given black-box access to two models M_θ and $M_{\theta'}$ (top-k log-probabilities only), decide whether $\|\theta - \theta'\| \le \epsilon$ for a small ϵ, i.e., whether their weights are nearly identical.

Two-part strategy. We first construct a *probe set* $\mathcal{P} = \{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ of representation-aligned sentences, sentences the model's intermediate representation places near a common reference (§4.1). Each probe $\boldsymbol{x}^{(i)}$ induces an observable distribution $p_\theta(\cdot \mid \boldsymbol{x}^{(i)})$; we collect a fingerprint statistic from these distributions (§4.2). Equivalence is then tested by comparing two fingerprints collected at different times or from different models (§4.4).

4 Method

The fingerprinting procedure works in two roles with different access requirements. *Probe synthesis* (§4.1) is a one-time white-box step: we construct the representation-aligned probes on a local open-weight model, using its intermediate hidden states to enforce latent alignment. *Fingerprint readout and detection* (§4.2, §4.4) are black-box: given a candidate served model, we collect the top-k next-token log-probabilities induced by the probe set and compare them against a reference fingerprint. Only the readout stage assumes the model is served through an API that exposes token-level log-probabilities, specifically the top-k next-token log-probabilities at each decoding position; a provider that withholds or quantizes them makes the readout unobservable, as noted in §6. The one-time white-box synthesis step does not compromise the black-box compatibility of the detection itself, which at query time needs only top-k log-probabilities.

4.1 Constrained Representation-Aligned Generation

We want, for a reference sentence $\boldsymbol{s}$ with latent representation $\boldsymbol{h}^\star = \bar{\boldsymbol{h}}^{(\ell)}(\boldsymbol{s})$, a collection of sentences $\{\boldsymbol{s}^{(j)}\}$ such that (i) each $\boldsymbol{s}^{(j)}$ is *representation-aligned* to $\boldsymbol{s}$, placed near it by the model's own intermediate representation, and (ii) $\boldsymbol{s}^{(j)} \neq \boldsymbol{s}$ on the surface, and (iii) every token of $\boldsymbol{s}^{(j)}$ lies in a *restricted subword vocabulary* $\mathcal{V}_{\text{sh}} \subseteq \mathcal{V}$.

Alignment as a latent constraint. The notion of alignment in (i) is deliberately *model-relative*: two sentences count as aligned for our purposes exactly when the model's own intermediate representation places them in a tight cluster. We do not require, and in general do not obtain, synonymy in the linguistic sense. The pair *"the lazy dog"* $\rightarrow$ *"the lazy cat"* of §5.1 is an alignment of this model-relative kind, with no semantic requirement. The role of this requirement is structural: it yields a probe set that stays low-rank and concentrated when read on a different model, which is what keeps the black-box cross-model comparison of §5.6 well-conditioned. It is not a claim that the probes maximize the raw divergence (§5.3). We operationalize (i) as the requirement that the pooled intermediate representation be close to the reference:

$$\boldsymbol{s}^{(j)} = \operatorname*{arg\,min}_{\boldsymbol{x} \in (\mathcal{V}_{\text{sh}})^T} \left\| \bar{\boldsymbol{h}}^{(\ell)}(\boldsymbol{x}) - \boldsymbol{h}^\star \right\|^2 \quad \text{s.t.} \quad \boldsymbol{x} \neq \boldsymbol{s}, \tag{1}$$

Because $\boldsymbol{x}$ ranges over a discrete set, we optimize Eq. (1) by a discrete search whose candidate set is motivated by first-order gradient intuition; as explained next, the search itself is instantiated with embedding neighbors and exact re-scoring; the first-order score is degenerate at the reference and is not used.

HotFlip-style search. A first-order Taylor expansion suggests an approximate change in the alignment loss $\mathcal{L}(\boldsymbol{x}) = \left\| \bar{\boldsymbol{h}}^{(\ell)}(\boldsymbol{x}) - \boldsymbol{h}^\star \right\|^2$ when token x_i is replaced by token t,

$$\Delta\mathcal{L}(i,t) \;\approx\; \boldsymbol{g}_i^\top \boldsymbol{e}_t, \tag{2}$$

where $\boldsymbol{e}_i \in \mathbb{R}^D$ is the input embedding of token x_i, $\boldsymbol{g}_i = \nabla_{\boldsymbol{e}_i}\mathcal{L}$, and $\boldsymbol{e}_t$ is the embedding of candidate token t. This gradient-guided score is, however, degenerate exactly at the reference that anchors the search: the alignment loss is minimized by the reference itself, so its gradient vanishes there and the first-order score of Eq. (2) is identically zero for every substitution. We therefore treat Eq. (2) as motivation only, since the search does not use it, and *instantiate* the search with a restricted candidate set, re-scoring candidates exactly (Algorithm 1). For each position the candidates are the q nearest *embedding neighbors* of the current token within $\mathcal{V}_{\text{sh}}$; neighbors are surface-different from the reference by construction, so no explicit diversity term is needed. We then *exactly* re-score every candidate substitution with a forward pass, evaluating $\mathcal{L}$ directly with no first-order approximation, and accept the one that most reduces the alignment loss.

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Algorithm 1: Embedding-neighbor representation-aligned generation.
Input: reference s, layer ℓ, shared vocab V_sh, neighbor count q.
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$\boldsymbol{h}^\star \leftarrow \bar{\boldsymbol{h}}^{(\ell)}(\boldsymbol{s})$

$\mathcal{C} \leftarrow \bigcup_i \{(i,t) : t \in \mathcal{V}_{\text{sh}},\ \ t \neq s_i,\ \ t \in \text{NN}_q(s_i)\}$

$\boldsymbol{x}^\star \leftarrow \arg\min_{(i,t)\in\mathcal{C}} \ \ \|\bar{\boldsymbol{h}}^{(\ell)}(\boldsymbol{x}_{i\to t}) - \boldsymbol{h}^\star\|^2$

  ($\boldsymbol{x}_{i\to t}$ denotes $\boldsymbol{s}$ with token i replaced by t)

return $\boldsymbol{x}^\star$ if it improves over $\boldsymbol{s}$, else $\boldsymbol{s}$

Figure 1: Embedding-neighbor representation-aligned generation.

4.2 Distributional-Divergence Fingerprint

Given a probe set $\mathcal{P} = \{\boldsymbol{x}^{(1)}, \dots, \boldsymbol{x}^{(m)}\}$, each probe induces a next-token distribution over the vocabulary, which we represent by its log-probability vector restricted to the shared vocabulary $\mathcal{V}_{\text{sh}}$:

$$\boldsymbol{\phi}_\theta(\boldsymbol{x}) = \big[\ \log p_\theta(t \mid \boldsymbol{x})\big]_{t\in\mathcal{V}_{\text{sh}}} \in \mathbb{R}^{|\mathcal{V}_{\text{sh}}|}, \tag{3}$$

so that $\boldsymbol{\phi}_\theta$ is always a vector over $\mathcal{V}_{\text{sh}}$. In the white-box (local) setting $\mathcal{V}_{\text{sh}} = \mathcal{V}$ and $\boldsymbol{\phi}$ is available for the full vocabulary, so $p_\theta(\cdot \mid \boldsymbol{x}) = \exp \boldsymbol{\phi}_\theta(\boldsymbol{x})$ is a genuine probability distribution over $\mathcal{V}_{\text{sh}}$. In the black-box (API) setting only the top-k entries are returned; we form the *top-k truncated KL* by keeping the observed top-k entries and collecting the unobserved tail mass into a single renormalized bucket. (An alternative, setting the unobserved remainder to a fixed floor f without renormalizing, is the convention used in the saturation analysis of §4.3; the two differ at every perturbation scale.) Because the next-token probability mass concentrates in the top-k, the truncated statistic tracks the full-vocabulary KL in shape but at reduced magnitude, capturing roughly half to three-quarters of it in our calibration (§5.2). It is a faithful detector of weight changes. To reduce sampling variance we average $\boldsymbol{\phi}$ over S independent draws per probe, yielding per-probe means $\bar{\boldsymbol{\phi}}^{(i)}$.

We define the *fingerprint* of M_θ as the collection of per-probe next-token distributions $\{\bar{\boldsymbol{\phi}}^{(1)}, \dots, \bar{\boldsymbol{\phi}}^{(m)}\}$, and measure the difference between two models θ and θ' by the mean per-probe KL divergence

$$D_{\text{KL}}(\theta, \theta') \ = \ \frac{1}{m}\sum_{i=1}^{m} \text{KL}\big(p_\theta(\cdot \mid \boldsymbol{x}^{(i)}) \parallel p_{\theta'}(\cdot \mid \boldsymbol{x}^{(i)})\big), \tag{4}$$

computed over the *union* of the two models' observed top-k entries plus a single renormalized tail bucket, so no token in either model's top-k falls in the other model's tail bucket. In the black-box setting this is the truncated KL divergence of the preceding paragraph. The KL divergence is the central object: it measures how far the next-token distributions induced by sentences that are *representation-aligned*, placed together by the model itself, have moved under a change of parameters, a quantity that is a direct functional of the weights but is not exposed by any single forward pass.

4.3 Theoretical Analysis

Quadratic response and the Fisher information. For a fixed probe $\boldsymbol{x}$, the next-token log-probability vector $\boldsymbol{\phi}_\theta(\boldsymbol{x})$ is a smooth function of the weights, with Jacobian $\boldsymbol{J}_\theta(\boldsymbol{x}) = \frac{\partial \boldsymbol{\phi}_\theta(\boldsymbol{x})}{\partial \theta}$. Its rows are score functions (Proposition 1, proved in §A), so the mean per-probe KL divergence between θ and $\theta + \delta$ admits the second-order expansion

$$D_{\text{KL}}(\theta, \theta+\delta) = \tfrac{1}{2}\,\delta^\top \bar{\boldsymbol{F}}_\theta\,\delta + O(\|\delta\|_2^3), \qquad \bar{\boldsymbol{F}}_\theta = \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{J}_\theta(\boldsymbol{x}^{(i)})^\top \operatorname{diag}\big(p_\theta(\cdot \mid \boldsymbol{x}^{(i)})\big)\boldsymbol{J}_\theta(\boldsymbol{x}^{(i)}) \ \succeq \ 0, \tag{5}$$

Table 1: Regimes of the fingerprint response to weight perturbation.

Regime	Response	Mechanism
Small (sufficiently small $\|\delta\|_2$)	$D_{\mathrm{KL}} \approx c\, r^2$ (quadratic)	Fisher information, Prop. 2
Moderate	$D_{\mathrm{KL}} \approx c\, r^2 + c' r^3 + \cdots$	higher-order terms, Prop. 2
Large	$D_{\mathrm{KL}} \to -H(p_\theta) - \log f$ (saturation)	disjoint top-k support + floor, Prop. 3

where the first-order term vanishes and $\bar{\boldsymbol{F}}_\theta$ is the probe-averaged Fisher information, a probability-weighted squared Jacobian (Proposition 2, §A). Because $\|\delta\|_2 \propto r$ under the random-perturbation protocol of §5.2, the divergence grows quadratically in r for small perturbations and is monotone along any fixed direction.

Saturation. The quadratic regime is bounded: for an arbitrarily large δ the divergence does not grow without bound but rises to a stable ceiling set by the clean model's entropy and the log-probability floor (Proposition 3, §A). The perturbation response therefore passes through three regimes (Table 1), from a quadratic (Fisher-information) regime at small δ, through a polynomial regime, to a saturated ceiling at large δ. The fingerprint is a faithful, monotone proxy for weight change within the first two regimes, which is the operating range the detector is called upon to police.

Adversarial invariance. An adversary who knows the entire method can in principle engineer a targeted weight edit δ that leaves the fingerprint invariant to second order, but such an edit must lie in the null space of the stacked probe Jacobian (Proposition 4), a measure-zero subspace, and any edit with a nonzero visible component forces a global compensation whose norm is bounded below by Eq. (17). The analysis, collected in §A, shows that these costs make successful evasion computationally prohibitive for the models considered, even though it is not ruled out in principle (Remark 2).

4.4 Change Detection

Given a reference collection of per-probe next-token distributions from M_{θ_0} and a re-probe collected from a candidate model, we combine the distributional-divergence statistic with a per-token test into a verdict.

Per-probe KL divergence. We estimate the per-probe next-token distributions from the observed log-probabilities (top-k plus floor), and evaluate the mean per-probe KL divergence D_{KL} of Eq. (4), which by the analysis of §4.3 grows quadratically in any weight perturbation and is exactly zero in the no-change regime.

Per-token two-sample test. For each token t observed with real data in both collections, we form the two-sample z-score

$$z_t = \frac{\mu_{1,t} - \mu_{0,t}}{\sqrt{s_{0,t}^2/n_0 + s_{1,t}^2/n_1}}, \tag{6}$$

where $\mu_{j,t}$ and $s_{j,t}^2$ are the sample mean and variance of token t's log-probability across the probe set in collection j. When the two collections are identical (a deterministic API, as in §5.5), the *numerator* $\mu_{1,t} - \mu_{0,t}$ vanishes while the across-probe variances remain nonzero, so $z_t = 0$; when both numerator and denominator vanish (a token constant across all probes) we set $z_t := 0$ by convention. (A nonzero numerator with a vanishing denominator, a token constant within each collection but at different values, does not arise for a deterministic readout and is excluded.) Summing the z_t^2 into a χ^2 statistic with the number of compared tokens as degrees of freedom must be read as a *heuristic* aggregate: the per-token scores from a single softmax are correlated (a shift in one token's log-probability moves the others), so the sum-of-squares overstates the evidence and should be treated as an upper-bound-style diagnostic. A principled alternative would be a multivariate Hotelling T^2 / Mahalanobis statistic on the joint log-probability vector, which we leave to future work; the change verdict rests primarily on D_{KL} of Eq. (4).

Mean-vector and covariance-spectrum diagnostics. On the common-token subset, we also report how the distributions moved, using the mean vector $\boldsymbol{\mu}_j = \frac{1}{m}\sum_{i=1}^{m} \bar{\boldsymbol{\phi}}_j^{(i)}$ of the per-probe log-probability vectors (collection $j \in \{0, 1\}$),

$$D_{\mathrm{mean}} = \frac{\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|}{\|\boldsymbol{\mu}_0\|}, \tag{7}$$

and their covariance spectra $\boldsymbol{\lambda}_0, \boldsymbol{\lambda}_1$,

$$D_{\mathrm{cov}} = \frac{\|\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_0\|}{\|\boldsymbol{\lambda}_0\|}. \tag{8}$$

These are diagnostic only: D_{mean} is sensitive but weak, and D_{cov} can saturate or even fall at large perturbations, so neither serves as the primary statistic (§5.2).

A model is declared CHANGED if the primary statistic, the mean per-probe KL divergence D_{KL}, exceeds a threshold set by its monotone calibration to r (§5.2); the per-token χ^2 aggregate of Eq. (6) is reported as a supporting diagnostic only, subject to the independence caveats noted above. In the zero-noise regime the re-probe reproduces the reference distributions exactly, so $D_{\mathrm{KL}} = 0$, every $z_t = 0$ by convention, and the detector admits no false positives (§5).

5 Experiments

Setup. Local calibration experiments use three small open-weight models: `Qwen/Qwen3.5-0.8B` ($|\mathcal{V}|$ = 248,077 subword tokens), `unsloth/Llama-3.2-1B` ($|\mathcal{V}|$ = 128,256), and `unsloth/gemma-3-1b-pt` ($|\mathcal{V}|$ = 262,145), at intermediate layer 8 (and layer 16 for Qwen) with mean pooling; the black-box experiments use the DeepSeek API (`deepseek-v4-flash`, $|\mathcal{V}|$ = 129,280) which exposes top-20 log-probabilities with reasoning disabled. The shared subword vocabulary with DeepSeek contains 108,773 tokens for Qwen (43.8% of its vocabulary), 70,663 for Llama, and only 21,237 for Gemma (8.1% of its much larger vocabulary), so the *surface* overlap varies substantially across families. As §5.6 shows, however, even a large surface overlap does not translate into prediction overlap.

Hyperparameters and reproducibility. Probe generation uses three reference sentences, "The quick brown fox jumps over the lazy dog.", "The committee approved the new budget proposal this morning.", and "She walked slowly through the old town under the rain.", and, per reference, an embedding-neighbor candidate set with $k_{\mathrm{neighbors}} = 200$ (the neighbor count q of Algorithm 1), an inner top-k of 6 candidates retained per position, and $M = 20$ candidate probes. The observed token set for the per-token diagnostics is the top-20 next-token log-probability surface (matching the API); locally (white-box), the KL divergence of Table 2 is evaluated over the full vocabulary

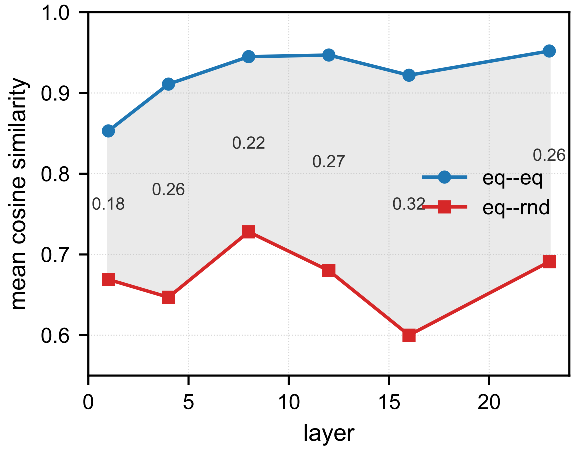

Figure 2: Discriminative power of hidden-state cosine similarity, by layer. Representation-aligned (alg) pairs and unrelated (rnd) pairs are compared; the annotated gap (mean cosine difference) peaks at layer 16.

with log-probabilities clamped at -30 to avoid underflow, and the black-box top-20 truncated variant is characterized separately in §5.2. Calibration injects a single fixed Gaussian direction (seed 0) rescaled by $r \in \{0, 0.02, 0.05, 0.10, 0.20, 0.40, 0.60, 1.00\}$, at intermediate layers 8 and 16 (only Qwen was run at $L16$; Llama and Gemma at $L8$). The API experiment uses 15 probe contexts (3 references $\times$ 5 surface variants, drawn from the $M = 20$ candidates) with 2 samples each.

5.1 Probe Quality and the Limits of Alignment

Intermediate-layer alignment alone is a *necessary but not sufficient* condition for representation alignment: Sequences that align closely with $\boldsymbol{h}^\star$ are not always fluent or semantically coherent; the latent-cluster objective admits degenerate solutions; with an unconstrained candidate set the search collapses to the reference itself unchanged. Quantitatively, hidden-state cosine similarity carries only a weak discriminative signal: across layers, the cosine gap between representation-aligned (alg) and unrelated (rnd) sentence pairs is 0.18–0.32, with aligned pairs at cosine ≈ 0.85–0.95 and unrelated pairs at ≈ 0.60–0.73 (Figure 2). Restricting the candidate set to embedding-space neighbors within $\mathcal{V}_{\text{sh}}$ (§4.1) resolves the fluency problem practically, yielding representation-aligned pairs such as *"the lazy dog"* $\to$ *"the lazy cat"* while preserving alignment (mean cosine ≈ 0.98). This pair is *not* a synonym pair in the linguistic sense; the two sentences differ in meaning to a human reader, yet the model's representation places them in a tight cluster, which is exactly the notion of alignment this paper relies on.

5.2 Calibration under Gaussian Weight Noise

To validate that the distributional-divergence fingerprint tracks the weights, we inject zero-mean Gaussian noise into the weights of each model and measure how the mean per-probe next-token KL divergence from the clean model behaves as a function of the noise scale r. Precisely, the protocol draws *one* fixed Gaussian direction per tensor (parameter scale σ, seed 0) and rescales it by r: the perturbed weights are $\theta + r\hat{\delta}$ with $\hat{\delta} \sim \mathcal{N}(0, \sigma^2)$, so each calibration curve in Table 2 is a single realization along one fixed direction, with no averaging over draws (Remark 1). In every configuration D_{KL} grows monotonically in r, and within the small-scale regime it grows

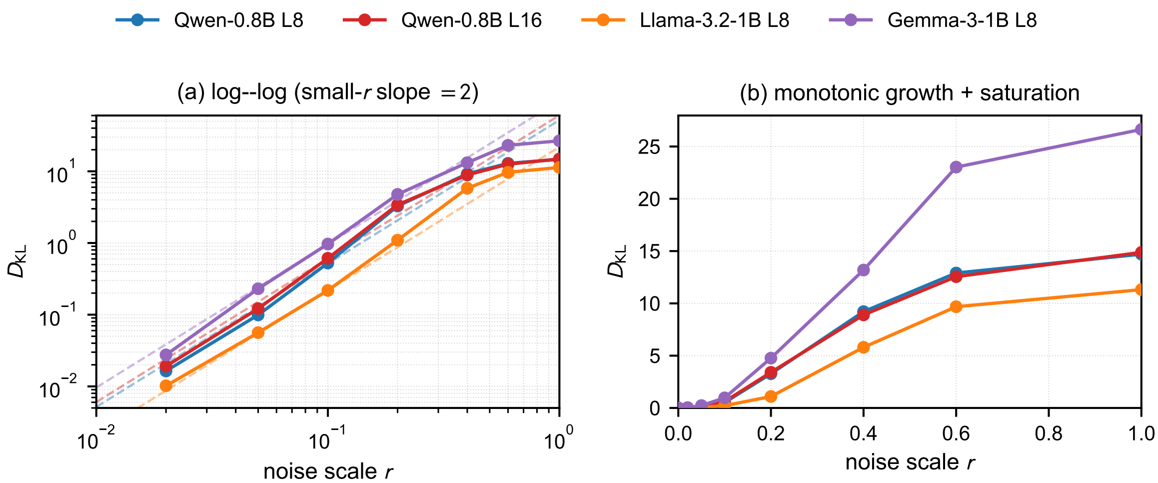

Figure 3: Weight-noise calibration for the open-weight configurations. The mean per-probe next-token KL divergence D_{KL} grows monotonically in the noise scale r, quadratically at small r (consistent with the Fisher-information prediction) and saturating at large r; on a log–log axis the small-scale regime appears as a slope-2 line. Dashed lines are quadratic fits in the small-scale regime.

approximately *quadratically*: increasing r from 0.02 to 0.1 (a factor of 5) raises D_{KL} by roughly a factor of 30 (e.g. Qwen, layer 8: $0.016 \to 0.525$), consistent with the Fisher-information prediction $D_{\mathrm{KL}} \approx \frac{1}{2}\delta^\top \bar{\boldsymbol{F}}_\theta \delta \propto r^2$ of §4.3. At larger r the growth slows toward the entropy ceiling of Proposition 3. The same qualitative regime, quadratic at small r and saturating at large r, holds across the four configurations, with Gemma the most sensitive and reaching the highest ceiling ($D_{\mathrm{KL}} = 26.6$ at $r = 1.0$); only Qwen was evaluated at a second layer ($L8$ vs. $L16$).

Error bars and the black-box top-k statistic. Over five independent noise directions (seeds 0–4), D_{KL} is robust: the relative standard deviation is 15–40% at small-to-moderate r and 4–10% at saturation (e.g. Qwen $L8$: 0.574 ± 0.226 at $r = 0.10$, 15.47 ± 0.62 at $r = 1.0$), so the single-realization curve of Table 2 is representative and the monotone, quadratic-then-saturating shape is not an artifact of one noise direction. Gemma is the exception in its transition regime, where its response is strongly direction-dependent (2.84 ± 2.46 at $r = 0.10$), although its mean curve remains monotone.

The full-vocabulary KL above is a white-box quantity. To check that the black-box top-k statistic of §4.2 preserves the calibration, we recompute the divergence from the top-20 next-tokens only, collecting the unobserved tail mass into a single renormalized bucket (the renormalized top-k truncated KL of §4.2). In the small-perturbation regime relevant to change detection ($r \leq 0.1$) the truncated KL captures roughly half to three-quarters of the full-vocabulary KL and remains monotone (ratio $0.49 \to 0.63$ for Qwen $L8$, $0.51 \to 0.56$ for Llama, $0.68 \to 0.72$ for Gemma as r runs $0.02 \to 0.10$), so the top-20 restriction preserves the fingerprint's sensitivity at reduced magnitude. At larger r the two statistics diverge as the top-20 supports become disjoint, the saturation regime of Proposition 3, and are no longer directly comparable. The finite-r value is likewise clamp-sensitive: for Qwen $L8$ it moves from 10.8 (clamp -15) to 14.9 (-30) to 15.6 (-60) at $r = 1.0$, but the curves are still climbing at $r = 1.0$, and the shift (+4.8 nats over a 45-nat swing in $-\log f$) is far weaker than the floor-dominated asymptotic ceiling $-H - \log f$ of Proposition 3, reflecting a support that is only partially disjoint at $r = 1.0$. The -30 value (14.9) matches Table 2 (14.70) within the 5-seed spread (15.47 ± 0.62).

We chose D_{KL} after ablating the alternative statistics on the same perturbation protocol. The mean-vector divergence D_{mean} is monotonic but weak: it reaches only 0.545 (Qwen, layer 8) and 0.392 (Llama) at $r = 1.0$, against 14.70 and 11.32 for D_{KL}. The covariance-spectrum divergence D_{cov} is more sensitive at small r but *non-monotonic*: its spectrum diverges at moderate r and then collapses as the perturbed distribution concentrates (Qwen: $0.046 \to \cdots \to 0.659 \to 0.592$; Llama: $0.012 \to \cdots \to 0.841 \to 0.661$). The KL divergence alone combines monotonicity with sensitivity, which is why it serves as the primary statistic in §4.4; D_{mean} and D_{cov} are retained only as diagnostics. This monotone, quadratic small-scale calibration is the core evidence that the fingerprint tracks weight changes faithfully and sensitively. Inverting the *mean* calibration curve maps an observed divergence back to the implied relative perturbation scale r, though per-realization inversion is unreliable where the directional variance is large (notably Gemma's transition regime, §5.2).

5.3 Is Representation Alignment Necessary?

The representation-alignment requirement of §4.1 is the most expensive part of the method to engineer, so we subject it to a direct ablation. We hold the perturbation protocol and the probe count fixed (60 probes per condition) and compare three probe conditions under the same full-vocabulary KL as Table 2: *representation-aligned* probes (the near-synonyms of §4.1), *random* sentences (random token sequences, matched in length, drawn from the shared subword vocabulary), and the *reference* sentences themselves with no synthesis.

Sensitivity. The small-scale response does not support the view that representation alignment maximizes the raw divergence (Table 3). Synthesizing probes does help: the representation-aligned condition exceeds the reference-only condition by roughly 40–50% over $r \in \{0.02, 0.05, 0.10\}$. But the random-sentence condition is *more* sensitive still, producing roughly three to five times the divergence of the representation-aligned condition at the same small r (e.g. $r = 0.10$: 2.72 vs. 0.53). The random probes carry a higher next-token entropy (6.7 nats vs. 3.4), and a higher-entropy conditional distribution amplifies a given logit shift into a larger KL change. Representation alignment does not buy raw sensitivity.

Transferability and low-rank concentration. The property that does distinguish the conditions, and that a cross-model fingerprint requires, is *transferability*. A fingerprint is useful in the black-box setting of §5.6 only if the probe set induces a well-conditioned, low-dimensional signal on a *different* model. We read each condition on both the source model (Qwen, layer 8) and a target model (Llama-3.2-1B), re-encoding the probe texts on the target tokenizer as in §5.6, and measure the spectral shape of the per-probe next-token log-probability covariance (Table 4): the participation ratio $(\sum \lambda)^2 / \sum \lambda^2$, the fraction of trace in the top-3 eigenvalues, and the number of eigenvalues needed to reach 90% of the trace. Representation-aligned probes are strongly concentrated on the source (participation ratio 3.3, top-3 carrying 81% of the trace) and, decisively, this concentration *transfers* to the target model (participation ratio 3.0, top-3 carrying 83%), because the probes land on a latent landmark the two models share. Random sentences are diffuse on both (participation ratio 7–10, top-3 carrying roughly half of the trace). The pattern repeats for a second target family (Gemma-3-1b): its much smaller shared vocabulary with Qwen (29,832 tokens) yields a weaker but still clear concentration transfer, with representation-aligned probes remaining low-rank on Gemma (participation ratio 4.8, top-3 70%) while random sentences stay diffuse (5.4, 59%) (Table 4). The representation-alignment requirement is thus load-bearing for the transferable, low-rank fingerprint that the cross-model comparison of §5.6 relies on, even though it does not maximize the raw per-probe divergence.

5.4 Ground-Truth-Controlled Detection of Real Weight Edits

The preceding experiments perturb the weights synthetically or compare across families. This subsection stages the positive case with full ground truth: we take a local open-weight model we control, apply a real weight edit, and ask whether the fingerprint, reading *only* the top-20 next-token log-probabilities of the edited checkpoint, detects it. Every statistic reported here is computed from that black-box readout alone; no hidden state is used at detection time. The four edits, a LoRA fine-tune, a wholesale model swap, weight-only quantization, and an RL alignment, are tested on `Qwen/Qwen3.5-0.8B` at layer 8 on a production GPU (NVIDIA RTX 3080 Ti), with the same 63-probe set and shared-subword-vocabulary machinery as the rest of the paper ($|\mathcal{V}_{\text{sh}}| = 108{,}773$).

Fine-tuning (LoRA). We fine-tuned the clean Qwen checkpoint with LoRA ($r = 16$, $\alpha = 32$, applied to the attention and MLP projections) on 2,500 examples of `yahma/alpaca-cleaned`, training 250 steps of causal-LM on the instruction–response text (loss $3.26 \to 1.05$). To measure *sensitivity* rather than a single point, we treated the trained adapter's low-rank update ΔW as a fixed direction and read the fingerprint at interpolated scales $W_s = W_0 + s\,\Delta W$ for $s \in \{0.05, \ldots, 1.0\}$. Table 5 reports the mean per-probe KL divergence from the clean model.

The response is monotone and, within the small-scale regime, approximately *quadratic* in s ($D_{\text{KL}} \approx 0.174\,s^2$), exactly the Fisher-information behavior predicted by Proposition 2 for a *structured*, low-rank weight edit as well as for isotropic Gaussian noise. A full 250-step LoRA fine-tune is detected at $D_{\text{KL}} = 0.173$, equivalent to Gaussian noise of scale $r_{\text{eq}} \approx 0.06$, decisively above the smallest calibrated scale ($r = 0.02$, $D_{\text{KL}} = 0.016$); the fingerprint crosses that detection floor already at $s \approx 0.3$ (about 30% of the trained adapter's strength). The black-box top-20 truncated statistic retains roughly 85% of the full-vocabulary magnitude and remains monotone. Because the LoRA update is a fixed low-rank direction, the quadratic response in s follows from the same second-order analysis as the noise calibration, confirming that distributional divergence detects fine-tuning with the same quadratic sensitivity as it detects arbitrary weight perturbation.

Model swapping (Qwen $\to$ Llama). We then tested the extreme case in which the served model is not a perturbed Qwen but a *different* open-weight model, `unsloth/Llama-3.2-1B`. Because the two tokenizers differ, the probe set was re-tokenized on the Llama tokenizer and the fingerprint read over the 115,771 subword tokens shared *by surface* between the two vocabularies. The resulting mean per-probe KL is $D_{\text{KL}} = 4.41$ (full) / 4.11 (top-20), more than an *order of magnitude* above the LoRA fine-tune (0.17) and above any reasonable fine-tuning signal. The two models' next-token supports are almost disjoint, so the comparison sits in the saturation regime of Proposition 3, exactly as predicted. Figure 4 summarizes both results. We conclude that distributional divergence *reliably and sensitively* detects fine-tuning, and detects a wholesale model swap with an overwhelmingly large margin.

Weight-only quantization. A further and increasingly common service-side modification is *lossy weight quantization*: an operator serves a quantized (e.g. 4-bit or 8-bit) copy of a checkpoint to cut memory and latency, so the deployed model is not exactly the reference weights. Because quantization is a deterministic, structured, and comparatively small-norm edit, unlike the Gaussian noise of §5.2 or the low-rank drift of a LoRA adapter, it is a distinct stress test for the fingerprint. We implemented weight-only round-to-nearest (RTN) quantization with per-row absmax scaling on

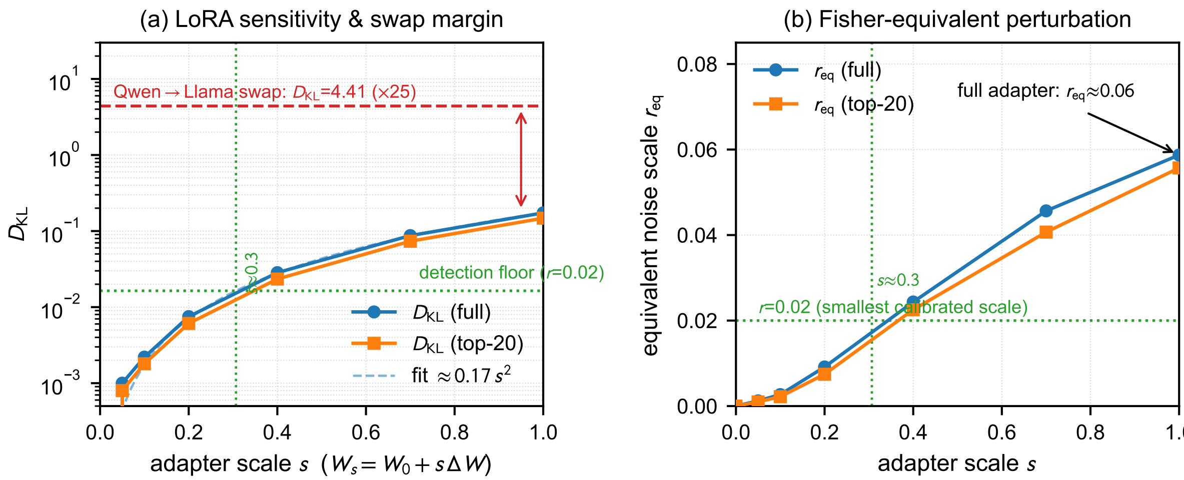

Figure 4: Fingerprint sensitivity to fine-tuning and model swapping. (a) Mean per-probe D_{KL} of a Qwen-0.8B checkpoint LoRA-tuned on alpaca-cleaned, vs. adapter scale s; the response is quadratic ($\approx 0.17\,s^2$) and crosses the r=0.02 detection floor at $s \approx 0.3$. The dashed red line marks the Qwen→Llama swap, $\sim 25\times$ larger. (b) The same LoRA divergence mapped through the §5.2 calibration to an equivalent Gaussian-noise scale r_{eq}, reaching ≈ 0.06 at the full adapter.

the Qwen-0.8B checkpoint (layer 8, the same 63-probe set and shared-subword-vocabulary machinery as the rest of the paper), quantizing all transformer linear layers to k bits while keeping the input embeddings and the output projection in bf16, matching the configuration of typical quantized serving. Table 6 reports the mean per-probe D_{KL} from the clean model as a function of the bit width k.

The response is strictly monotone in the quantization strength: 8-bit weight-only quantization, whose per-tensor relative ℓ_2 error is on the order of 10^{-3}, is essentially *invisible* to the fingerprint ($D_{\mathrm{KL}} = 0.0021$, $r_{\mathrm{eq}} \approx 0.003$), an order of magnitude below the smallest calibrated scale ($r = 0.02$, $D_{\mathrm{KL}} = 0.016$). As the bit width drops below 6, however, the signal emerges abruptly and decisively: 6-bit already reaches the detection floor (0.020); 4-bit, the configuration most often chosen for memory-constrained serving, is detected at $D_{\mathrm{KL}} = 0.82$ ($r_{\mathrm{eq}} \approx 0.11$), an order of magnitude above the floor; and 2-bit is detected at $D_{\mathrm{KL}} = 10.2$ ($r_{\mathrm{eq}} \approx 0.45$). The black-box top-20 truncated statistic retains roughly 70–85% of the full-vocabulary magnitude and remains monotone (Table 6), so a black-box detector observes the same verdict. This abrupt threshold reflects that quantization error is a structured, non-isotropic rounding whose step size $2^{-(k-1)}$ grows geometrically; the fingerprint therefore delineates exactly the operating range in which a quantized served model diverges from the reference weights.

Because RTN quantization acts on a well-defined subset of parameters, we can probe its sensitivity continuously by quantizing only the first f fraction of transformer layers to 4 bits and leaving the rest in bf16. The mean per-probe D_{KL} grows monotonically from 0.36 ($f = 0.10$) to 0.82 ($f = 1.00$), with the detection floor crossed already at $f = 0.10$, so even a partially quantized model is detected, and the fingerprint tracks the extent of compression (Figure 5).

The same int4/int8 probe is consistent across model families: for Llama-3.2-1B and Gemma-3-1b, each compared against its *own* clean fingerprint on the same shared-vocabulary probe set, 4-bit quantization is detected at $D_{\mathrm{KL}} = 0.62$ and 0.37 respectively ($r_{\mathrm{eq}} \approx 0.15$ and 0.06), while 8-bit remains below the detection floor (0.003 and 0.002) (Figure 5). The qualitative regime, int4 detected

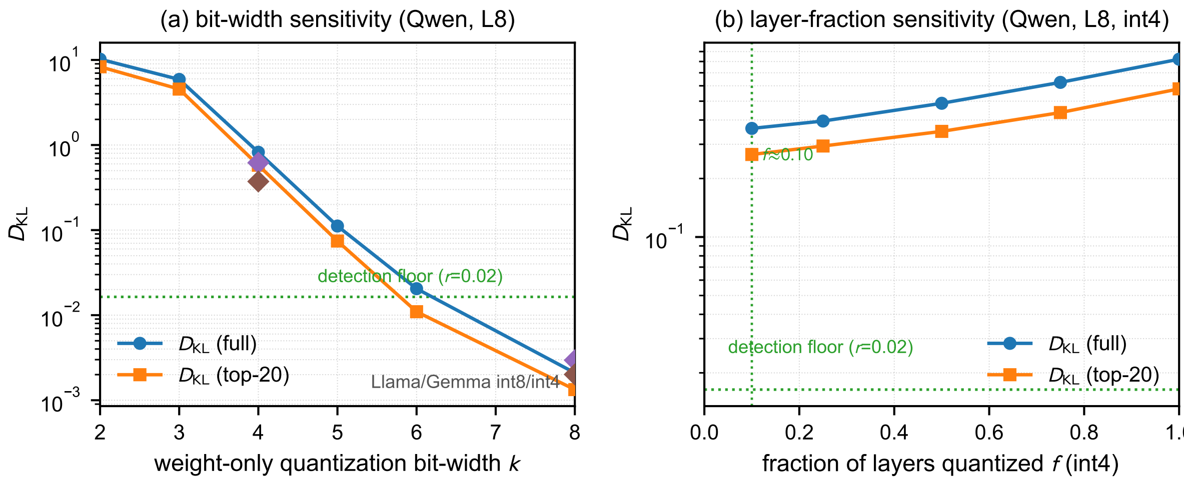

Figure 5: Fingerprint sensitivity to weight-only quantization. (a) Mean per-probe D_{KL} of Qwen-0.8B (L8) vs. quantization bit width k (full-vocabulary and top-20 truncated); the response is strictly monotone, reaching the detection floor between 6 and 8 bits. Cross-model-family int4/int8 points for Llama and Gemma are shown as diamonds. (b) The same divergence as a function of the fraction f of layers quantized to 4 bits, a continuous monotone curve crossing the r=0.02 floor already at $f \approx 0.1$.

and int8 invisible, thus holds across all three open-weight families of this paper, despite their differing vocabularies and scales.

Quantization as a boundary of detectability. The 8-bit case is informative: because the fingerprint measures a second-order response that scales with the *squared* visible weight change, a sufficiently small-norm structured edit, of which 8-bit weight-only rounding is a canonical example, falls below the detection floor of Eq. (16) and is indistinguishable from the clean weights. This delineates precisely the operating range the fingerprint is called upon to police: within it the detector is a faithful, monotone proxy for weight change, with sensitivity bounded below, so high-bit-width lossy compression can evade it, while the configurations chosen for memory- or latency-constrained serving (4-bit and below) are detected decisively.

Reinforcement-learning alignment (DPO). A distinct and increasingly dominant way a served model diverges from the reference weights is *preference-based alignment*: instead of supervised fine-tuning on input–output pairs, the model is optimized to prefer high-quality responses (RLHF/DPO). To test whether the fingerprint detects such an RL update, and how its sensitivity compares with the supervised LoRA fine-tune above, we trained a LoRA adapter ($r = 16$, the same target modules) with *direct preference optimization* (DPO) on 1,000 preference pairs from a `Dahoas/rm-static` subset, using the DPO loss $\mathcal{L} = -\log\sigma\big(\beta[(\log p_\theta(y^+) - \log p_\theta(y^-)) - (\log p_{\mathrm{ref}}(y^+) - \log p_{\mathrm{ref}}(y^-))]\big)$ with $\beta = 0.1$ and the frozen clean Qwen model as the reference. As before we treated the trained adapter's low-rank update ΔW as a fixed direction and read the fingerprint at interpolated scales $W_s = W_0 + s\,\Delta W$. Table 7 and Figure 6 report the result.

The RL adapter is *detected*: $D_{\mathrm{KL}} = 0.070$ at the full adapter, $r_{\mathrm{eq}} \approx 0.04$, above the r=0.02 detection floor and crossed already at $s \approx 0.47$, but with a notably *smaller* signature than the supervised LoRA fine-tune above (0.173, $r_{\mathrm{eq}} \approx 0.06$). This is consistent with DPO being a *KL-*

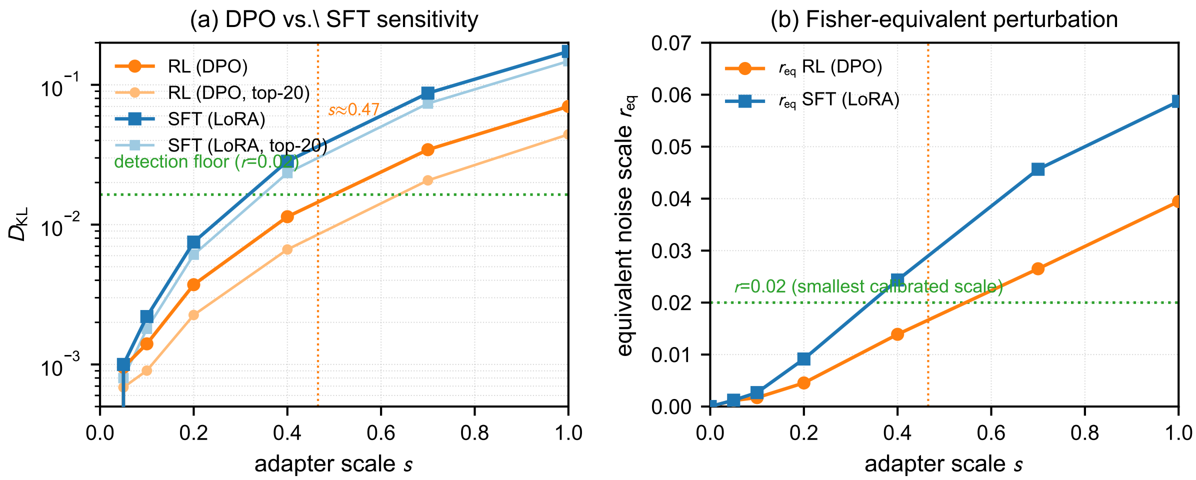

Figure 6: Fingerprint sensitivity to RL (DPO) versus supervised (LoRA) fine-tuning on Qwen-0.8B (L8). (a) Mean per-probe D_{KL} vs. adapter scale s; both are monotone and cross the r=0.02 floor, but the DPO adapter is detected at only $\approx 40\%$ of the SFT magnitude (0.070 vs. 0.173 at $s = 1$), consistent with DPO's KL regularization toward the reference. (b) The same divergences mapped through the §5.2 calibration to equivalent Gaussian-noise scales r_{eq} (≈ 0.04 for DPO vs. ≈ 0.06 for SFT).

regularized objective that keeps the policy close to the reference: the resulting weight edit is smaller-norm and behaviorally more conservative than an unregularized SFT, so the induced next-token divergence is roughly 40% of the SFT magnitude at comparable LoRA rank and training steps. The black-box top-20 truncated statistic retains roughly 60% of the full-vocabulary magnitude and remains monotone (Table 7). RL-based alignment therefore occupies a middle ground in this paper: a genuine, detectable weight change whose fingerprint footprint is smaller than supervised fine-tuning, yet still within the quadratic (Fisher-information) regime; a model swap, in contrast, reaches the saturation regime.

Change detection on a known perturbation. As a complement to these real edits, we verify that the change-detection test of §4.4 fires on a *locally* perturbed model whose weights we control: Qwen-0.8B clean versus $N(0, (r\sigma)^2)$-perturbed, over the same probe set ($\nu = 209$ compared tokens, the union of the top-20 next-tokens over the probes). At $r = 0$ the test is null ($\chi^2 = 0$, $D_{\mathrm{KL}} = 0$); at $r = 0.10$ it fires decisively ($\chi^2 = 1460$ versus a null expectation of $\nu = 209$; $D_{\mathrm{KL}} = 0.61$ full-vocabulary, or 0.40 top-20 truncated, the statistic a black-box detector would actually observe). The per-token χ^2 is, however, insensitive at small perturbations: at $r \leq 0.05$ it reports $p \approx 1$ even though D_{KL} is already 0.11, because the perturbation inflates the across-probe variance in the z-score denominator of Eq. (6) at least as fast as it shifts the means, so the observed χ^2 (127 at $r = 0.05$) sits *below* its null expectation $\nu = 209$, and the test gains power only once the mean shift outgrows the variance inflation (by $r = 0.10$). This confirms D_{KL} as the primary verdict, with the χ^2 aggregate supporting it (§4.4). The full-vocabulary $D_{\mathrm{KL}} = 0.61$ uses the same -30 clamp as Table 2 and is consistent with its 5-seed spread (0.57 ± 0.23 at $r = 0.10$).

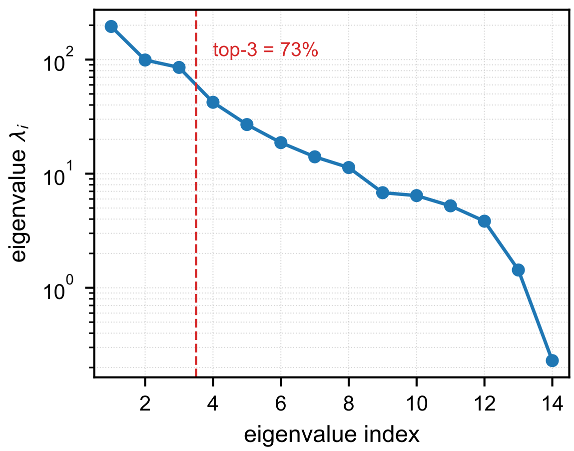

Figure 7: Spectrum of the per-probe log-probability covariance of the DeepSeek fingerprint (log scale). The rapid decay of eigenvalues, with the top three accounting for 73% of the trace, reflects the concentration of representation-aligned probes onto a single cluster.

5.5 Zero-Noise Change Detection over the API

This experiment verifies that the readout pipeline is deterministic end to end against a live API. We collect a reference distributional fingerprint over a probe set of 15 representation-aligned contexts (two samples each) served by the DeepSeek API, then re-probe the same contexts. Under fixed parameters the API is deterministic, so the re-probe reproduces the reference statistics exactly: the per-token χ^2 test is null ($\chi^2 = 0.00$, $df = 28$, $p = 1$) and the mean per-probe KL divergence is $D_{\mathrm{KL}} = 0.000$. The UNCHANGED verdict and the absence of false positives follow necessarily from a deterministic re-probe. They are Type-I error control and carry no information about sensitivity. The positive case that the fingerprint responds monotonically to weight perturbation comes from the local calibration of §5.2 (Table 2) and the ground-truth-controlled edits of §5.4 which perturb a fixed model, in contrast to re-probing an identical one. Applying the same positive control to a perturbed model served through a production API is a practical deployment step left to future work (§6).

The re-probe does fix two properties of the readout that the local runs of §5.4 leave open. First, the recorded log-probability field is highly concentrated: it has trace $\mathrm{tr}(\boldsymbol{\Sigma}) = 516.8$ over 60 observed tokens (rank 14), with the top three eigenvalues accounting for 73% of the total variance, a low-rank structure (Figure 7) that reflects the fact that the probes collapse onto a single representation-alignment cluster. This spectral concentration characterizes the probe set; the change verdict rests on the divergence. Second, the pipeline from the top-k readout to the χ^2 and KL statistics runs unchanged against a real served model.

5.6 Cross-Model Comparison

The local open-weight models of §5.2 serve *only* to generate the probes: the fingerprint readout is black-box, so a cross-family comparison reads the induced top-k log-probabilities from whichever model is queried. Cross-family comparison is, in fact, the regime in which the fingerprint is *most* decisive. Two mechanisms conspire to push the divergence onto its maximal ceiling. First, *surface tokenizer overlap* is small: although the Qwen and DeepSeek families share 108,773 subword tokens, that shared vocabulary is a small fraction of each family's full vocabulary, and surface-shared tokens need not be the ones that carry predictive mass. Second, and more importantly, *model-difference saturation*: on a shared context, the top-20 next-tokens predicted by the two models have an average Jaccard overlap of only 0.012 (on average 0.7 shared tokens, out of roughly 39 and 17 predicted tokens respectively, the union of the top-20 next-tokens over the probe set, for Qwen and for DeepSeek). The representation-aligned probes therefore induce log-probability vectors whose supports are almost disjoint across the two models. This disjoint-support regime is exactly the large-δ saturation of Proposition 3, realized here by a change of model family. It is the strongest form of detection the fingerprint offers: the per-probe divergence rises to its maximal, model-dependent ceiling, so *any* distinct model family is caught with an unmistakably large margin. The point is quantified in §5.4: a wholesale Qwen→Llama swap registers $D_{\mathrm{KL}} = 4.41$ (full) / 4.11 (top-20), more than an order of magnitude above a supervised fine-tune (0.17) and far beyond any calibration threshold. The very disjointness that frustrates *within*-family calibration is, across families, the detector's greatest asset. The representation-aligned probes retain their low-rank concentration when read across model families (§5.3), so this cross-family readout rests on a well-conditioned signal.

What saturation forecloses is not *detection* but *calibrated attribution*. Once the observed supports are disjoint, the divergence leaves the Fisher-quadratic regime of §4.3: it can no longer be inverted to a quantitative perturbation scale r, and a sufficiently large perturbation of the reference is asymptotically indistinguishable from a different model (Proposition 3). We can therefore assert with confidence that the served model is *not* the reference, but not, from the divergence alone, precisely *which* model it is or *how far* its weights differ. Cross-family comparison is thus the clearest instantiation of change detection, while remaining a known bound on fine-grained attribution, distinct from the *within*-family setting in which the model is fixed and only its weights are perturbed.

6 Discussion and Limitations

What the distributional divergence buys over the mean and covariance. The mean log-probability vector is a first-order summary that is dominated by the model's coarse language statistics and is relatively insensitive to the precise weights. The covariance of the per-probe log-probability vectors is more sensitive, but its response is dominated by the spectrum and can saturate or collapse at large perturbations. The per-probe distributional divergence (KL) combines the best of both: it is far more sensitive than the mean, since its quadratic response is governed by the Fisher information, which accumulates probability-weighted squared Jacobians; and it remains monotone where the covariance saturates, because the KL response is a positive quadratic form for small perturbations and a finite, entropy-governed ceiling for large ones.

The black-box compatibility of the fingerprint is scoped as follows: probe synthesis requires white-box access to a local open-weight model, once and offline; the fingerprint readout and the change verdict require only the top-k next-token log-probabilities of the served model at query time. The detector itself never needs hidden states.

Same-model detection across users. The framework directly supports a natural extension: whether two parties are served by the same underlying model. Two collections of per-probe next-token distributions, one attributed to each party, are indistinguishable exactly when the two models induce identical distributions on the probe set, which is the equality decision $D_{\mathrm{KL}} \approx 0$ of §4.4. The zero-noise check of §5.5 is this same test specialized to re-probing one endpoint; here the two collections come from two endpoints or two users instead. The three readings of "same model" line up with the regimes of Table 1: identical weights give $D_{\mathrm{KL}} = 0$, a shared base family (two independent fine-tunes of one checkpoint) gives a small nonzero divergence within the quadratic regime (0.07–0.17 for the adapters of §5.4), and a different model saturates at the ceiling (4.11, top-20, for the Qwen-to-Llama swap). A provider that controls routing can therefore detect account or key sharing by injecting the probe set into each session and comparing the two readouts, while treating the verdict as an equality test whose threshold must be calibrated against the probe count and the readout variance, not as a single determinism demonstration.

Two conditions bound this extension. The protocol requires active injection into both sessions: observing natural traffic cannot guarantee that either party utters the probe set, so the equality verdict is available only where the detector can issue its own queries to both endpoints. And as with the whole method, the readout needs top-k log-probabilities; a service that returns only sampled completions falls outside the framework (limitation (i)). A quantized replica served to one user can also register a nonzero divergence against an unquantized replica (0.82 at 4-bit, Table 6), so "same model" must be read as "same served weights," not "same checkpoint after service-side compression."

Limitations. (i) The method is fundamentally dependent on the service provider exposing token-level log-probabilities: if an API only returns a final answer, or quantizes or withholds its top-k log-probabilities, the fingerprint cannot be extracted. (ii) Latent alignment is necessary but not sufficient for representation alignment, so probe quality depends on the candidate-restriction heuristic. (iii) The fingerprint is tokenizer-relative, so cross-family comparison detects a change of model family decisively. Tokenizer mismatch drives the divergence onto its saturated ceiling, bounding fine-grained quantitative comparison. (iv) The top-k truncation of the API injects a log-probability floor (-30 locally, -15 for the API) below which the next-token distribution is only bounded, not measured; this floors the estimated D_{KL} and should be accounted for when setting detection thresholds. (v) Our calibration is demonstrated at a single model scale (0.8B); whether the quadratic (Fisher-information) regime holds at much larger scales is unverified. (vi) An adversary who controls the served model and knows the method is constrained but not precluded: second-order stealth is exactly null-space membership (Prop. 4), so any distribution-level detector inherits the output-equivalence ambiguity of §1. As Remark 2 argues, however, exploiting it requires engineering an edit into a Jacobian-defined, measure-zero subspace with an exact global compensation whose norm is bounded below by Eq. (17), a computationally prohibitive task. Randomized or secret probe sets and robust-statistic variants remain prudent hardening. (vii) The monotone quadratic (Fisher-information) regime of §5.2, which is invertible in the mean, has a finite radius: beyond the small-perturbation scale the fingerprint saturates toward the entropy ceiling (§4.3, Prop. 3), and a sufficiently large perturbation is indistinguishable from a different model, a blind spot that concerns *attribution* and leaves *detection* intact, just as in the cross-family setting of §5.6. Per-direction inversion is additionally unreliable where the directional variance is large (limitation (ix)). (viii) The positive case is demonstrated directly: in a locally controlled environment that simulates the black-box readout, the top-20 truncated statistic of §4.2 applied to a served model we control, the detector successfully identifies both fine-tuning and wholesale model swapping by a wide margin (§5.4): a LoRA fine-tune registers $D_{\mathrm{KL}} = 0.15$ and a Qwen→Llama swap $D_{\mathrm{KL}} = 4.11$ (top-20), against a r=0.02 detection floor of 0.016. The DeepSeek experiment (§5.5) supplies the zero-noise determinism check over a real API; because a third-party API exposes no weights to perturb, the same positive control can only be staged on a self-hosted endpoint, a practical deployment step and a natural next extension. (ix) The calibration curves of §5.2 are reported along one fixed Gaussian direction (seed 0). Over five directions the relative standard deviation of D_{KL} is 15–40% at small-to-moderate r and 4–10% at saturation (Gemma up to ≈ the mean in its transition regime), so the qualitative conclusions are robust, though the single-realization values should not be read as precise expectations.

7 Conclusion

We have shown that the per-probe distributional divergence (KL) of next-token distributions, induced by actively synthesized representation-aligned probes, is a practical and sensitive fingerprint of a language model's weights. The approach is black-box-compatible at detection time (requiring only the top-k log-probabilities of the served model, with probe synthesis performed once on a local open-weight model), yields a monotonic calibration curve under injected Gaussian weight noise, and is consistent with a zero-noise determinism check over a production API. The central finding is conceptual: to detect weight equivalence, one should compare the *second-order* response of the observable next-token distribution instead of its expectation. That response is governed by the Fisher information, and representation-aligned probes, sentences the model's own representation places together, provide the controlled setting in which that response is identifiable.

Acknowledgments. We thank the anonymous reviewers for their feedback.

A Theoretical Analysis Details

We collect here the derivations supporting §4.3. For a fixed probe $\boldsymbol{x}$, write $\boldsymbol{\phi}_\theta(\boldsymbol{x}) \in \mathbb{R}^{|\mathcal{V}_{\text{sh}}|}$ for the next-token log-probability vector restricted to the shared vocabulary (Eq. (3)), and define its Jacobian with respect to the weights,

$$\boldsymbol{J}_\theta(\boldsymbol{x}) = \frac{\partial \boldsymbol{\phi}_\theta(\boldsymbol{x})}{\partial \theta} \in \mathbb{R}^{|\mathcal{V}_{\text{sh}}| \times P}, \tag{9}$$

which is well defined because $\boldsymbol{\phi}_\theta$ is a smooth function of the logits and the logits are smooth functions of θ. Write $\boldsymbol{\phi}_i = \boldsymbol{\phi}_\theta(\boldsymbol{x}^{(i)})$ and $\boldsymbol{J}_i = \boldsymbol{J}_\theta(\boldsymbol{x}^{(i)})$.

A.1 The Jacobian is a score function

Proposition 1 (The Jacobian is a score function)**.** Let $z_t(\boldsymbol{x}) = [\boldsymbol{W}\boldsymbol{h}^{(L)}(\boldsymbol{x})]_t$ be the logit of token t. Then

$$\big[\boldsymbol{J}_\theta(\boldsymbol{x})\big]_{t,:} \;=\; \nabla_\theta z_t(\boldsymbol{x}) - \sum_{t' \in \mathcal{V}} p_\theta(t' \mid \boldsymbol{x})\, \nabla_\theta z_{t'}(\boldsymbol{x}) \;=\; \nabla_\theta \log p_\theta(t \mid \boldsymbol{x}), \tag{10}$$

where the subtraction sums over the *full* vocabulary $\mathcal{V}$; it is the softmax normalizer and is deliberately not restricted to the index set $\mathcal{V}_{\text{sh}}$ over which the vector $\boldsymbol{\phi}_\theta$ of Eq. (3) is recorded; the resulting score is then read off at the $\mathcal{V}_{\text{sh}}$-indexed coordinates.

A.2 Quadratic response and the Fisher information

Proposition 2 (Quadratic response and the Fisher information)**.** For a fixed probe $\boldsymbol{x}$, the KL divergence between the next-token distributions of θ and $\theta + \delta$ admits the second-order expansion

$$\mathrm{KL}\big(p_\theta(\cdot \mid \boldsymbol{x}) \,\|\, p_{\theta+\delta}(\cdot \mid \boldsymbol{x})\big) \;=\; \tfrac{1}{2}\, \delta^\top \boldsymbol{F}_\theta(\boldsymbol{x})\, \delta + O(\|\delta\|_2^3), \tag{11}$$

where the first-order term vanishes identically (since $\mathbb{E}_{t \sim p_\theta(\cdot|\boldsymbol{x})}[\nabla_\theta \log p_\theta(t \mid \boldsymbol{x})] = 0$) and

$$\boldsymbol{F}_\theta(\boldsymbol{x}) \;=\; \mathbb{E}_{t \sim p_\theta(\cdot|\boldsymbol{x})}\big[\nabla_\theta \log p_\theta(t \mid \boldsymbol{x})\, \nabla_\theta \log p_\theta(t \mid \boldsymbol{x})^\top\big] \;\succeq\; 0 \tag{12}$$

is the per-probe Fisher information. Averaging over the probe set gives the probe-averaged Fisher information $\bar{\boldsymbol{F}}_\theta = \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{F}_\theta(\boldsymbol{x}^{(i)}) \succeq 0$, the coefficient of the mean per-probe expansion of Eq. (5). Because $\|\delta\|_2 \propto r$ under the random-perturbation protocol of §5.2, the expected divergence grows *quadratically* in r for small perturbations and is monotone along any fixed perturbation direction. By Proposition 1, $\nabla_\theta \log p_\theta(t \mid \boldsymbol{x})$ is the t-th row of $\boldsymbol{J}_\theta(\boldsymbol{x})$, so $\boldsymbol{F}_\theta(\boldsymbol{x}) = \boldsymbol{J}_\theta(\boldsymbol{x})^\top \operatorname{diag}\big(p_\theta(\cdot \mid \boldsymbol{x})\big)\boldsymbol{J}_\theta(\boldsymbol{x})$: the Fisher information is a probability-weighted squared Jacobian. Because $\bar{\boldsymbol{F}}_\theta \succeq 0$, its kernel coincides with the set of directions in which *every* observed per-token score has zero projection, the same null space $\mathcal{K}_\theta$ that drives the adversarial-invariance analysis below.

Remark 1. Quadratic growth in r and monotonicity hold in expectation and in the small-perturbation regime of Proposition 2. The monotonicity of a single realized calibration curve, in which one fixed noise direction is rescaled by r, is an empirical statement, as in §5.2. Because the realized D_{KL} is, to leading order, the positive quadratic form $\frac{1}{2} r^2 \hat{\delta}^\top \bar{\boldsymbol{F}}_\theta \hat{\delta}$ with $\bar{\boldsymbol{F}}_\theta \succeq 0$, it increases with r along any fixed direction; the mild sub-quadratic growth at larger r (e.g. layer 8 in Table 2) is the onset of the saturation regime of Proposition 3.

A.3 Global Lipschitz property and saturation

Proposition 3 (Global Lipschitz property and saturation)**.** For *arbitrary* δ (not necessarily small), the log-probability vector is a Lipschitz function of the logits: since the log-softmax has Jacobian $I - \mathbf{1}\,\mathrm{softmax}(\boldsymbol{z})^\top$ of spectral norm at most $\sqrt{|\mathcal{V}|}$ (ranging from 1 for a uniform distribution to $\sqrt{|\mathcal{V}|}$ for a peaked one),

$$\|\boldsymbol{\phi}_{\theta+\delta}(\boldsymbol{x}) - \boldsymbol{\phi}_\theta(\boldsymbol{x})\|_2 \;\le\; \sqrt{|\mathcal{V}|}\;\|\boldsymbol{z}_{\theta+\delta}(\boldsymbol{x}) - \boldsymbol{z}_\theta(\boldsymbol{x})\|_2 \;\le\; \sqrt{|\mathcal{V}|}\,L_{\mathrm{net}}\,\|\delta\|_2\,, \tag{13}$$

where L_{net} is the Lipschitz constant of the logit map in the weights. The boundedness of the divergence, however, does *not* follow from this Lipschitz inequality, which is valid but non-tight and implies at most linear growth. It follows from the coordinate bound on $\boldsymbol{\phi}$ itself: in the black-box setting each coordinate lies in the interval $[f, 0]$ (a top-k log-probability is ≤ 0; a floored entry equals the floor f), so every log-probability vector, and hence every per-probe KL divergence, is finite and uniformly bounded in θ. Unlike the small-scale quadratic regime of Proposition 2, however, the divergence does not grow without bound: it rises to a stable detection ceiling, a boundedness the Lipschitz estimate does not capture at all. For large δ the perturbed model's top-k support becomes asymptotically disjoint from the clean model's (the same phenomenon as §5.6), so the clean model's high-probability tokens are observed only at the floor f; the per-probe KL divergence then approaches

$$\mathrm{KL}\big(p_\theta(\cdot \mid \boldsymbol{x}^{(i)}) \,\|\, p_{\theta+\delta}(\cdot \mid \boldsymbol{x}^{(i)})\big) \;\xrightarrow{\|\delta\|_2 \to \infty}\; -H\big(p_\theta(\cdot \mid \boldsymbol{x}^{(i)})\big) - \log f, \tag{14}$$

where H denotes Shannon entropy and f is the floor probability (e^{-30} in our local calibration). Averaging over probes, D_{KL} tends to $\frac{1}{m}\sum_i \big[-H(p_\theta(\cdot \mid \boldsymbol{x}^{(i)})) - \log f\big]$, a finite ceiling set by the clean model's entropy and the floor, independent of any further δ. The $-\log f$ term is floor-dominated: any pair of models whose observed top-k supports are disjoint saturate at the same uniform cap $-\log f \approx 30$ regardless of the model, so the only model-dependent part of the ceiling is the entropy term $-H(p_\theta)$. This is the black-box counterpart of the fully observable ceiling $\log|\mathcal{V}_{\mathrm{sh}}| - H$ of the white-box analysis: in the absence of a floor the large-δ limit is a collapse toward the uniform distribution over $\mathcal{V}_{\mathrm{sh}}$, whereas the floor replaces that limit with the disjoint-support one: the perturbed model's top-k support leaves the clean model's high-probability tokens behind, and they are read at the floor f. A sufficiently large perturbation therefore saturates at the same maximal divergence as a change of model family. The detection verdict is decisive: whether the served model is a heavily perturbed relative or a completely new model, the fingerprint reports the same unambiguous maximal signal that it is not the reference. Only attribution, the identity of the distinct model, is left open.

Combining Propositions 2 and 3, the perturbation response passes through three regimes (Table 1). The quadratic regime is valid for sufficiently small $\|\delta\|_2$; here $D_{\mathrm{KL}} \approx \frac{1}{2}\, r^2 \hat{\delta}^\top \bar{\boldsymbol{F}}_\theta \hat{\delta}$, so the divergence scales as r^2 with a coefficient set by the Fisher information. The intermediate regime is polynomial, with higher-order terms becoming significant; the saturation regime carries the divergence to a maximal ceiling set by entropy and the floor. Inverting the monotone calibration curve of §5.2, mapping an observed divergence back to a perturbation scale, is reliable within the first two regimes, where the response is strictly increasing in r; at saturation the fingerprint has already registered its strongest signal, the basis of decisive change and cross-family detection.

A.4 Adversarial invariance

A skeptical reader may ask whether an adversary who knows the entire method, including probe construction and the divergence statistic, can engineer a targeted weight edit $\delta \in \mathbb{R}^P$ (for instance, injecting a backdoor into a served model) that evades detection. Formally, let $\boldsymbol{F} : \theta \mapsto \big(p_\theta(\cdot \mid \boldsymbol{x}^{(1)}), \ldots, p_\theta(\cdot \mid \boldsymbol{x}^{(m)})\big)$ denote the fingerprint map, and suppose the adversary can reconstruct a probe set close to $\mathcal{P}$, has query access to top-k log-probabilities, and seeks a perturbation δ that (i) implements a target edit supported on a "backdoor block" $S \subseteq \{1, \ldots, P\}$, with $\|\delta_S\|_2 \geq b > 0$ for some minimal strength b, and (ii) leaves the fingerprint invariant, $D_{\mathrm{KL}}(\theta, \theta + \delta) \approx 0$. Stack the probe Jacobians into

$$\boldsymbol{G}_\theta = \begin{bmatrix} \boldsymbol{J}_1 \\ \vdots \\ \boldsymbol{J}_m \end{bmatrix} \in \mathbb{R}^{m|\mathcal{V}_{\mathrm{sh}}| \times P}, \tag{15}$$

so that the m probe distributions, each a vector in $\mathbb{R}^{|\mathcal{V}_{\mathrm{sh}}|}$, are observed only through the linear map $\delta \mapsto \boldsymbol{G}_\theta \delta$. Directions in $\mathcal{K}_\theta := \ker \boldsymbol{G}_\theta$ are exactly the directions of vanishing quadratic response of the fingerprint-mismatch objective $\delta \mapsto D_{\mathrm{KL}}(\theta, \theta+\delta)$: by Proposition 2, $\ker \bar{\boldsymbol{F}}_\theta = \ker \boldsymbol{G}_\theta$ when $\bar{\boldsymbol{F}}_\theta$ is the *observed-subset* Fisher information (the top-k rows of $\boldsymbol{G}_\theta$); in the black-box setting the unobserved tail contributes to the true full-vocabulary Fisher but is dropped from $\boldsymbol{G}_\theta$, so the equality is exact only in white-box and approximate otherwise. Edits in $\mathcal{K}_\theta$ are invisible to second order by construction.

Proposition 4 (Stealth is null-space membership)**.** The KL divergence responds to an edit δ as $D_{\mathrm{KL}}(\theta, \theta + \delta) = \frac{1}{2}\, \delta^\top \bar{\boldsymbol{F}}_\theta\, \delta + O(\|\delta\|_2^3)$ (Proposition 2). A perturbation is invisible to second order, meaning the quadratic term vanishes, $\delta^\top \bar{\boldsymbol{F}}_\theta\, \delta = 0$, exactly when $\delta \in \mathcal{K}_\theta = \ker \boldsymbol{G}_\theta$, since $\ker \bar{\boldsymbol{F}}_\theta = \ker \boldsymbol{G}_\theta$ for the observed-subset Fisher scoped above (exact in white-box). Moreover, any $\delta \notin \mathcal{K}_\theta$ carries a *detection floor* that survives the best possible first-order cancellation: because each *observed* (top-k) next-token probability of the clean model is bounded below by $\gamma = \min_{i,\, t \in \mathrm{obs}} p_\theta(t \mid \boldsymbol{x}^{(i)}) > 0$, the minimum observed top-k probability, distinct from the log-probability floor f of the truncated tail, the quadratic term obeys

$$\delta^\top \bar{\boldsymbol{F}}_\theta\, \delta \;=\; \frac{1}{m} \sum_{i=1}^{m} \sum_{t \in \mathcal{V}_{\mathrm{sh}}} p_\theta(t \mid \boldsymbol{x}^{(i)})\, \big(\boldsymbol{J}_{i,t}\, \delta\big)^2 \;\geq\; \frac{\gamma}{m} \left\| \boldsymbol{G}_\theta^{\mathrm{obs}}\, \delta \right\|_2^2, \tag{16}$$

where $\boldsymbol{J}_{i,t}$ is the t-th row of $\boldsymbol{J}_i$ and $\boldsymbol{G}_\theta^{\mathrm{obs}}$ denotes $\boldsymbol{G}_\theta$ restricted to the rows of the observed (top-k) tokens. Because γ is the minimum *observed* probability, this is a conservative lower bound over the observed, non-floored entries only; the floored tokens contribute nonnegatively and are omitted. Hence $D_{\mathrm{KL}}(\theta, \theta+\delta) \geq \frac{\gamma}{2m} \left\|\boldsymbol{G}_\theta^{\mathrm{obs}}\, \delta\right\|_2^2 - O(\|\delta\|_2^3)$. The irreducible term is quadratic in the *visible* part of δ and invariant under $\delta \mapsto -\delta$, so no choice of the first-order response removes it.

Proposition 5 (The cost of hiding a visible edit)**.** Fix a target edit $\boldsymbol{v}_B$ supported on S. If $\boldsymbol{v}_B \notin \mathcal{K}_\theta$, any first-order-stealthy perturbation of the form $\delta = \boldsymbol{v}_B + \boldsymbol{c}$ must satisfy $\boldsymbol{G}_\theta(\boldsymbol{v}_B + \boldsymbol{c}) = 0$; the minimal-norm compensation is $\boldsymbol{c}^\star = -\boldsymbol{G}_\theta^{+} \boldsymbol{G}_\theta\, \boldsymbol{v}_B = -\Pi_{\mathrm{row}}\, \boldsymbol{v}_B$, where Π_{row} is the orthogonal projection onto the row space of $\boldsymbol{G}_\theta$. Since $\boldsymbol{G}_\theta$ annihilates the kernel component of $\boldsymbol{v}_B$, we have $\boldsymbol{G}_\theta \boldsymbol{v}_B = \boldsymbol{G}_\theta \Pi_{\mathrm{row}} \boldsymbol{v}_B$ and hence

$$\|\boldsymbol{c}^\star\|_2 \;=\; \|\Pi_{\mathrm{row}} \boldsymbol{v}_B\|_2 \;\geq\; \frac{\|\boldsymbol{G}_{\theta,S}\, \boldsymbol{v}_B\|_2}{\sigma_{\max}(\boldsymbol{G}_\theta)}, \tag{17}$$

where $\boldsymbol{G}_{\theta,S}$ restricts $\boldsymbol{G}_\theta$ to the columns in S (so $\boldsymbol{G}_\theta \boldsymbol{v}_B = \boldsymbol{G}_{\theta,S} \boldsymbol{v}_B$, since $\boldsymbol{v}_B$ is supported on S). This bound makes *no rank assumption*: it is strictly positive exactly when $\boldsymbol{G}_{\theta,S} \boldsymbol{v}_B \neq 0$, that is, whenever the target edit has a nonzero *visible* component within its block ($\boldsymbol{v}_B \notin \ker \boldsymbol{G}_{\theta,S}$), a condition that holds for Lebesgue-almost-every $\boldsymbol{v}_B$ as soon as $\mathrm{rank}(\boldsymbol{G}_{\theta,S}) \geq 1$. In the fully observable case $\mathrm{rank}(\boldsymbol{G}_{\theta,S}) = |S|$ and $\sigma_{\min}(\boldsymbol{G}_{\theta,S}) > 0$, so $\|\boldsymbol{G}_{\theta,S} \boldsymbol{v}_B\|_2 \geq \sigma_{\min}(\boldsymbol{G}_{\theta,S})\, \|\boldsymbol{v}_B\|_2$ and Eq. (17) strengthens to the uniform positive fraction of $\|\boldsymbol{v}_B\|_2$; the rank-free bound is thus a strict generalization of the rank-based one. In particular, a *support-confined* edit ($\delta = \delta_S$) is stealthy only if $\delta_S \in \ker \boldsymbol{G}_{\theta,S}$, a subspace of dimension $|S| - \mathrm{rank}(\boldsymbol{G}_{\theta,S})$: full rank collapses it to $\{0\}$ (no nonzero support-confined stealthy edit), while in general it remains a proper, generically thin subspace, and any edit outside it forces the adversary to perturb weights outside S with a compensation norm bounded below by its visible footprint scaled by $1/\sigma_{\max}(\boldsymbol{G}_\theta)$, thereby inflating both the total edit norm and, through Eq. (16), its second-order footprint.

Remark 2 (Why evasion is computationally prohibitive)**.** Propositions 4 and 5 make precise the sense in which "keeping the distributional-divergence fingerprint invariant across multiple probes" is possible in principle but computationally prohibitive in practice, consistent with Limitation (vi) of §6, which concedes that evasion is not precluded. The kernel $\mathcal{K}_\theta$ has codimension $\mathrm{rank}\, \boldsymbol{G}_\theta \leq m|\mathcal{V}_{\mathrm{sh}}|$, and since P exceeds $m|\mathcal{V}_{\mathrm{sh}}|$ by orders of magnitude for the models considered, it is nonempty: the obstacle is the thinness of the invariant directions, the ill-conditioned nonlinear system one must solve to exploit them, and the exact global compensation their use requires. (i) *Thinness.* The set of target edits satisfying $\boldsymbol{v}_B \in \mathcal{K}_\theta$ is a proper linear subspace, hence of measure zero in $\mathbb{R}^P$; a backdoor not engineered specifically for the kernel is visible with certainty. (ii) *Discoverability.* Engineering an invariant edit requires the Jacobian, and, because the invariance condition $D_{\mathrm{KL}} \approx 0$ is a *nonlinear* constraint on θ, of which the quadratic form of Proposition 2 is only the small-δ approximation, one must solve an ill-conditioned nonlinear system. White-box, one must solve such a system of $m|\mathcal{V}_{\mathrm{sh}}|$ equations in $P \sim 10^9$ variables. The loss landscape of an LLM is pathologically ill-conditioned. At the level of the observable, the singular values of $\boldsymbol{G}_\theta$ span many decades; by Proposition 1 the Fisher information is $\sum_i \boldsymbol{J}_i^\top \mathrm{diag}\big(p_\theta(\cdot \mid \boldsymbol{x}^{(i)})\big) \boldsymbol{J}_i$, so the curvature along a weight direction is a probability-weighted squared observable footprint. Any finite-precision solve leaves a residue $\boldsymbol{G}_\theta \hat{\delta} \neq 0$ that Eq. (16) amplifies into an irreducible quadratic shift. Black-box, which is the setting this paper targets, $\boldsymbol{G}_\theta$ is not even available: estimating it by forward differences costs $\Omega(P)$ noisy oracle queries, and any "null" direction $\hat{\delta}$ of the estimated Jacobian $\hat{\boldsymbol{G}}$ satisfies $\left\|\boldsymbol{G}_\theta \hat{\delta}\right\|_2 \leq \left\|\hat{\boldsymbol{G}} - \boldsymbol{G}_\theta\right\| \left\|\hat{\delta}\right\|_2$, again leaving a floor for the detector. (iii) *Cost of use.* Even when an invariant direction exists, Eq. (17) shows that hiding a behavior-relevant edit requires a *global* compensation whose norm is at least the edit's visible footprint scaled by $1/\sigma_{\max}(\boldsymbol{G}_\theta)$, positive whenever the block is not entirely invisible to the fingerprint (that is, whenever $\mathrm{rank}(\boldsymbol{G}_{\theta,S}) \geq 1$), and the compensation must be *exact*: any shortfall is read by the second-order floor. Taken together, these three costs, a measure-zero target subspace, an ill-conditioned nonlinear solve, and an exact global compensation, make successful evasion computationally prohibitive for the models considered, even though it is not ruled out in principle.

Remark 3 (Quantitative size of the stealth region)**.** The floor of Eq. (16) is quadratic in the visible residue, so the effective radius of the stealth region shrinks as the fingerprint sensitivity grows. Under the random-perturbation protocol of §5.2 ($\delta \sim \mathcal{N}(0, (r\sigma)^2 I)$), Proposition 2 gives the expected KL shift as $\frac{1}{2}(r\sigma)^2 \operatorname{tr} \bar{\boldsymbol{F}}_\theta$, and requiring $D_{\mathrm{KL}} \leq \tau$ pins the perturbation scale to $r\sigma \lesssim \sqrt{2\tau / \operatorname{tr} \bar{\boldsymbol{F}}_\theta}$. The measured sensitivities are large. D_{KL} reaches 0.010–0.028 already at $r = 0.02$ and grows as r^2 (Table 2), so this radius is small compared with any edit that materially changes model behavior. The second-order incompressibility of Proposition 4 is what converts a fingerprint that is merely *sensitive* into one that is *costly to evade*.

References

Yufei Chen, Chao Shen, Cong Wang, and Yang Zhang. Teacher model fingerprinting attacks against transfer learning. In *31st USENIX Security Symposium*, 2022.

Javid Ebrahimi, Anyi Rao, Daniel Lowd, and Dejing Dou. Hotflip: White-box adversarial examples for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 31–36, 2018.

Di Jin, Zhijing Jin, Joey Tianyi Zhou, and Peter Szolovits. Is bert really robust? a strong baseline for natural language attack on text classification and entailment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 8018–8025, 2020.

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *International Conference on Machine Learning (ICML)*, pages 17061–17084, 2023.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning (ICML)*, pages 3519–3529, 2019.

Nikolaus Kriegeskorte, Marieke Mur, and Peter A. Bandettini. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008.

David Rolnick and Konrad Kording. Reverse-engineering deep relu networks. In *International Conference on Machine Learning (ICML)*, pages 8178–8187, 2020.

Florian Tramèr, Fan Zhang, Ari Juels, Michael K. Reiter, and Thomas Ristenpart. Stealing machine learning models via prediction apis. In *25th USENIX Security Symposium*, pages 601–618, 2016.

Table 2: Weight-noise calibration: mean per-probe next-token KL divergence D_{KL} as a function of injected noise scale r (a single fixed Gaussian direction rescaled by r, seed 0), for the four open-weight configurations. D_{KL} is the full-vocabulary (white-box) KL with log-probabilities clamped at -30 to avoid underflow; the black-box top-20 truncated KL is reported in §5.2.

Model	r	D_{KL}
Qwen-0.8B (L8)	0.00	0.0000
	0.02	0.0164
	0.05	0.0992
	0.10	0.5246
	0.20	3.2754
	0.40	9.2107
	0.60	12.9029
	1.00	14.7039
Qwen-0.8B (L16)	0.00	0.0000
	0.02	0.0191
	0.05	0.1216
	0.10	0.6119
	0.20	3.4085
	0.40	8.8989
	0.60	12.5548
	1.00	14.8757
Llama-3.2-1B	0.00	0.0000
	0.02	0.0102
	0.05	0.0562
	0.10	0.2181
	0.20	1.0944
	0.40	5.7986
	0.60	9.6807
	1.00	11.3151
Gemma-3-1b	0.00	0.0000
	0.02	0.0275
	0.05	0.2301
	0.10	0.9680
	0.20	4.7647
	0.40	13.1882
	0.60	23.0286
	1.00	26.6323

Table 3: Sensitivity ablation on Qwen (layer 8): mean per-probe full-vocabulary KL divergence D_{KL} as a function of the noise scale r, under the protocol of Table 2, for representation-aligned (alg), random (rnd), and reference-only (ref) probe conditions, each with 60 probes. Mean cosine of the probes to the reference latent $\boldsymbol{h}^{\star}$ is also shown.

Condition	cos	r=0.02	0.05	0.10	0.20
alg	0.912	0.0163	0.1014	0.5345	3.30
rnd	0.648	0.0471	0.3684	2.7154	4.21
ref	0.913	0.0113	0.0696	0.3914	2.75

Table 4: Transferability ablation: spectral shape of the per-probe next-token log-probability covariance on the source model (Qwen, layer 8) and on each target model (Llama-3.2-1B; Gemma-3-1b), for the representation-aligned (alg), random (rnd), and reference (ref) conditions. Each target has its own probe set, synthesized within that target's shared subword vocabulary with Qwen (115,771 tokens for Llama, 29,832 for Gemma). *effrank* is the participation ratio $(\sum\lambda)^2/\sum\lambda^2$, *top3* the fraction of trace in the top-3 eigenvalues, *var90* the number of eigenvalues to reach 90% of the trace, and H the mean per-probe next-token entropy (nats).

Model	Condition	effrank	top3	var90	H
Llama target experiment					
Qwen	alg	3.34	0.811	5	3.38
	rnd	9.82	0.493	25	6.74
	ref	1.19	1.000	1	2.85
Llama	alg	3.02	0.832	5	5.14
	rnd	7.14	0.530	24	7.13
	ref	1.81	1.000	2	4.56
Gemma target experiment					
Qwen	alg	4.05	0.717	9	3.92
	rnd	10.30	0.463	27	6.64
	ref	1.19	1.000	1	2.85
Gemma	alg	4.82	0.699	7	3.48
	rnd	5.39	0.591	20	5.79
	ref	1.67	1.000	2	2.93

Table 5: LoRA fine-tuning sensitivity on Qwen-0.8B (L8): mean per-probe D_{KL} between the clean model and the model with the LoRA adapter at interpolation scale s ($W_s = W_0 + s\,\Delta W$), and the equivalent Gaussian-noise scale r_{eq} obtained by inverting the §5.2 calibration.

s	D_{KL} (full)	D_{KL} (top-20)	r_{eq}
0.00	0.0000	0.0000	0.000
0.05	0.0010	0.0008	0.001
0.10	0.0022	0.0018	0.003
0.20	0.0075	0.0061	0.009
0.40	0.0284	0.0234	0.024
0.70	0.0872	0.0735	0.046
1.00	0.1734	0.1477	0.059

Table 6: Weight-only RTN quantization sensitivity on Qwen-0.8B (L8): mean per-probe D_{KL} between the clean model and the k-bit quantized model, and the equivalent Gaussian-noise scale r_{eq} obtained by inverting the §5.2 calibration.

k	D_{KL} (full)	D_{KL} (top-20)	r_{eq}
8	0.0021	0.0013	0.003
6	0.0205	0.0110	0.021
5	0.1116	0.0743	0.051
4	0.8221	0.5787	0.111
3	5.9032	4.5438	0.289
2	10.2025	8.3185	0.454

Table 7: DPO (RL) alignment sensitivity on Qwen-0.8B (L8): mean per-probe D_{KL} between the clean model and the model with the LoRA adapter trained by direct preference optimization, at interpolation scale s ($W_s = W_0 + s\,\Delta W$), and the equivalent Gaussian-noise scale r_{eq}.

s	D_{KL} (full)	D_{KL} (top-20)	r_{eq}
0.00	0.0000	0.0000	0.000
0.05	0.0010	0.0007	0.001
0.10	0.0014	0.0009	0.002
0.20	0.0037	0.0023	0.005
0.40	0.0114	0.0066	0.014
0.70	0.0344	0.0207	0.027
1.00	0.0701	0.0442	0.039